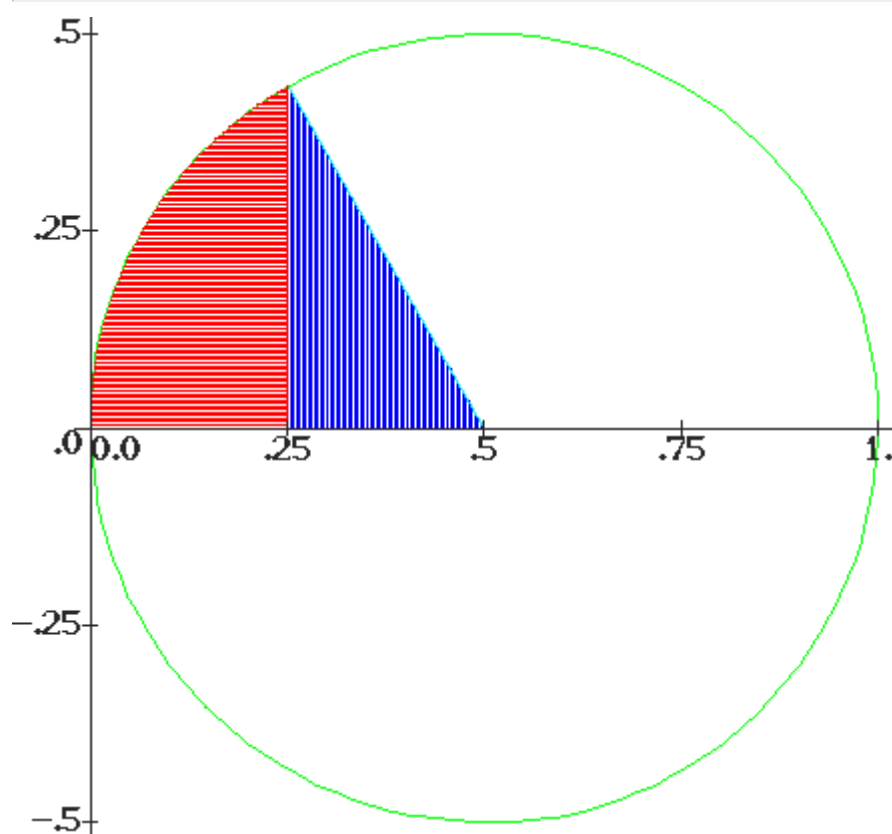




Additional Discussion of Newton's formula



It is possible to show the expression $\pi = \frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{x - x^2} dx$ in a rather informal way with the help of the diagram above.

Claim : The definite integral $\int_0^{\frac{1}{4}} \sqrt{x - x^2} dx$ represents the blue area in the above circle.

Let $y = \sqrt{x - x^2}$, then

$$y^2 = x - x^2 \quad (1)$$

$$x^2 - x + y^2 = 0 \quad (2)$$

$$\left(x - \frac{1}{2}\right)^2 + y^2 = \left(\frac{1}{2}\right)^2 \quad (3)$$

which is the locus of the above circle with radius $r = \frac{1}{2}$ unit and center at $\left(\frac{1}{2}, 0\right)$.

It follows that the definite integral from $x=0$ to $x = \frac{1}{4}$ of the function is equal to the blue area in the circle shown above. Claim: $\frac{1}{24} \cdot \frac{3\sqrt{3}}{4}$ is the area of the red triangle within the circle.

$$\text{Area of triangle} = \frac{1}{2} \cdot \text{base} \cdot \text{height} \quad (4)$$

$$= \frac{1}{2} \cdot \frac{1}{2} \cos\left(\frac{\pi}{6}\right) \cdot \frac{1}{2} \sin\left(\frac{\pi}{6}\right) \quad (5)$$

$$= \frac{1}{24} \cdot \left(3 \frac{\sqrt{3}}{4}\right) \quad (6)$$

Observation : The blue area + the red area is one sixth the area of the circle.

Deduction:

$$\frac{\pi \left(\frac{1}{2}\right)^2}{6} = \frac{1}{24} \cdot \frac{3\sqrt{3}}{4} + \int_0^{\frac{1}{4}} \sqrt{x - x^2} dx \quad (7)$$

$$\pi = \frac{3\sqrt{3}}{4} + 24 \int_0^{\frac{1}{4}} \sqrt{x - x^2} dx \quad (8)$$
